MILITARY SPECIFICATION

STARTER,FIRE,NP3,M2

This Amendment forms a part of Military Specification MIL-S-13175H(AR), dated 15 May 1984, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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3.5: Compliance of Chemical Mixture.

Delete in its entirety and substitute: " For each chemical mixture an analysis shall be performed as specified in para. 4.5.5."

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Add the following new paragraph:

4.5.5, Chemical Analysis.

4.5.5.1, Scratcher Mixture." The following procedure is to be employed in analyzing the mixture Phosphorous,Red (MIL-P-211, Class C) Sand-Polystyrene. This prodecure is to be applied before solubilizing the binder during processing. Constituent materials are to be in compliance with the applicable specification and drawing. Alternate procedures for this analysis are acceptable only if approved by AMSMC-QAT-I, Dover, N.J. 07801-5001.

1. Polystyrene. Transfer an accurately weighed portion of approximately two grams of the sample to a tared 30 ml. medium porosity fritted class thimble crucible. Using Soxhlet or Roweg-extraction apparatus and toluene as the solvent, separate the Polystyrene from the other constituents.

FSC 1370

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Completeness of extraction will be indicated by the absence of Polystyrene pellets and should be substantiated by constant values obtained for the weight of the thimble crucible, and contents as the extraction procedure continues. For this purpose remove the thimble crucible from the extraction apparatus and dry it in an oven for 45 minutes at 80 ± 5° C.

$$\% \text{ Polystyrene} = \frac{A-B}{W} \times 100$$

Where: A = Weight of sample and crucible before extraction

B = Weight of sample and crucible after extraction

W = Weight of sample

2. Sand. Transfer the contents of the thimble crucible to a 400 ml. beaker using approximately 10 ml. of water. Add 35 ml. of concentrated nitric acid. Cover the beaker with a watch glass, warm on a hot plate until vigorous reaction stops and then boil down to a volume of approximately 20 ml. Add 10 ml. of concentrated perchloric acid and boil down to a volume of 3 or 4 ml. After cooling, transfer the residue in the beaker to a tared 30 ml. medium porosity fritted glass crucible. Using small portions of water to effect a quantitative removal, a wash the contents of the crucible with at least ten 5 ml. portions of water, at room temperature, using a glass stirring rod to assure good contact with all portions of the residue. Aspirate between additions of the wash water and finally aspirate to dryness. Place crucible, with residue, in a drying oven maintained at 110 ± 5° C. for a minimum of sixty minutes. Cool the crucible,and residue in a desiccator and obtain its weight.

$$\% \text{ Sand} = \frac{C-D}{W} \times 100$$

Where: C = Weight of residue and crucible

D = Weight of crucible

W = Weight of sample, from Section 1.

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3. Phosphorus Red.

% Phosphorus, Red = 100 - (E+F)

Where: E = % Polystyrene

F = % Sand

4.5.5.2, Match Head Mixture. The following procedure is to be employed in analyzing the mixture Potassium Chlorate (MIL-P-150, Grade B, Class 2), Antimony Sulfide (MIL-A-159, Class 1), Polystyrene. This procedure is to be applied before solubilizing the binder during processing, Constituent materials are to be in compliance with the applicable specification and drawing. Alternative procedures for this analysis are acceptable only if approved by AMSMC-QAT-I,Dover N.J. 07801-5001.

1. Potassium Chlorate. Transfer on accurately weighed portion of approximately one gram of the sample to a tared 30 ml. medium porosity fritted glass crucible. To the material in the crucible, add five 10 ml. portions of water at room temperature. Maintain each portion of the water in contact with the sample for approximately three minutes using a glass stirring rod for mixing. Aspirate after each addition of water. Place the crucible, with residue, in a drying oven maintained at 110 $\pm$ 5°C. for a minimum of sixty minutes. Cool the crucible, with residue, in a desiccator and obtain its weight.

$$\% \text{ Potassium Chlorate} = \frac{A-B}{W} \times 100$$

Where: A = Weight of sample and crucible before extraction.

B = Weight of residue and crucible after extraction.

W = Weight of sample

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2. Antimony Sulfide. To the crucible containing the residue, obtained as indicated in Section 1, add six 15 ml. portions of concentrated hydrochloric acid at approximately 80°C. Maintain each portion of acid in contact with the contents of the crucible for approximately three minutes using a glass stirring rod for mixing. Aspirate after each addition of acid. The disappearance of all dark material indicates complete reaction of the antimony sulfide. Wash the contents of the crucible with four 5 ml. portions of 2% hydrochloric acid and finally with four 5 ml. portions of water, both at room temperature. Aspirate thoroughly and place crucible, with contents, in a drying oven maintained at 110 ± 5° C. for a minimum of 60 minutes. Cool the crucible, and residue, in a desiccator and obtain its weight.

$$\% \text{ Antimony Sulfide} = \frac{B-C}{W} \times 100$$

Where: B = Weight of residue and crucible, from Section 1.

C = Weight of residue and crucible after acid extraction

W = Weight of sample, from Section 1.

3. Polystyrene.

$$\% \text{ Polystyrene} = 100 - (D+E)$$

Where: D = % Potassium Chlorate

E = % Antimony Sulfide."

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